

Homework 10: Optimization

EEC 3150

For each Python exercise, write a separate Python script. Submit all scripts on Blackboard under the appropriate assignment. Name your script files with the following format:

HW<assignment number>_Ex<exercise
number>_<FirstInitial><LastName>.py

Example: HW01_Ex01_GWest.py

For short answer or math exercises, either do your work on a sheet of paper and upload a picture of it to Blackboard or submit some kind of text file or Word document to Blackboard. You can place all such exercises in a single document as long as you clearly label each exercise. These files should follow a similar naming convention as the Python scripts.

Exercise 1 (25 points, Python)

Consider the function

$$f(x, y) = (x^2 + y^2)^{0.5} - 0.6(\cos(x) + \cos(y))$$

1. Implement and run a 2D gradient descent method to find the minimum of the function
2. On a single 3D plot...
 - (a) Plot the function in 3D over the domain $[-15, 15] \times [-15, 15]$ using `plot_surface`
 - (b) Plot the path of the GD method on the xy plane beneath the surface

Exercise 2 (25 points, Python)

Redo the previous exercise with 2D simulated annealing. You will need to specify the covariance matrix of the multivariate normal distribution. Use something like

```
width**2*np.eye((2,2))
```

where `width` is the proposal width used in the same sense as 1D simulated annealing.

Exercise 3 (25 points, Python)

We can represent the wavefunction of the electron in a hydrogen atom as

$$\psi(r) = \frac{2}{R^{3/2}} \exp(-r/R)$$

The energy of this wavefunction is a function of the R parameter

$$E(R) = 2B \left(\frac{A^2}{2R^2} - \frac{A}{R} \right)$$

where

$$A = \hbar^2 / (kme^2)$$
$$B = mk^2e^3 / (2\hbar^2)$$

and

- $\hbar$ (reduced Planck constant) = 1.05×10^{-34} J s
- k_e (Coulomb constant) = 8.98×10^9 N m²/C²
- e (electron charge) = 1.60×10^{-19} C
- m (electron mass) = 9.10×10^{-31} kg

The value of R which minimizes this energy is the Bohr radius and the minimum value of E is the ground state energy of the hydrogen atom.

1. Implement and run simulated annealing to determine the value of the Bohr radius and the ground state energy
2. Print the values
3. TIP: the Bohr radius is approximately 10^{-10} m so be aware of this when setting the value of your proposal width